Carnegie Mellon University
Computer Science Academy

Design and Development Guide

Before beginning this project, be sure to review the rubric to ensure you include all the required elements.

Description

Describe your program. What do you envision your app will do?

Note: At this step, do not be concerned about how to do it, just focus on what you envision.

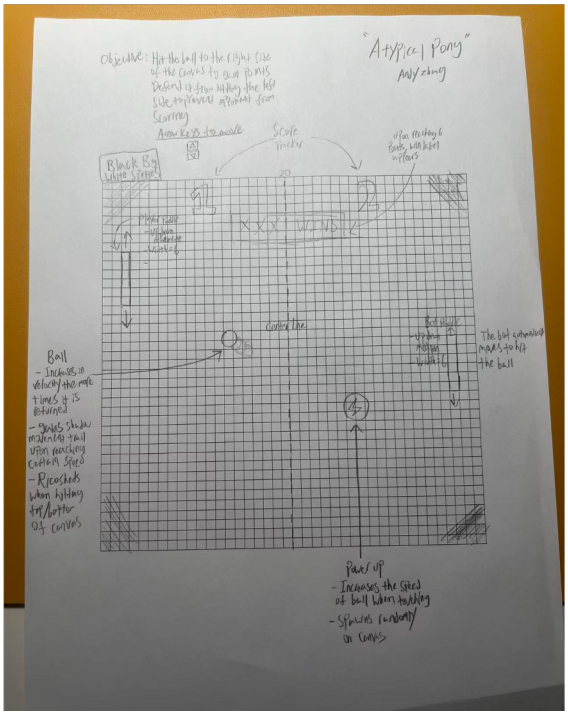
I want to create a more interesting version of the classic game Pong. The player will move their paddle up and down to hit the ball, whereas a bot will try to defend it. The game will follow the rules of Classic Pong, except the ball accelerates when struck, making it significantly more challenging to defend the more returns the ball gets. Power-ups also regularly spawn randomly on the canvas and will drastically increase the speed of the ball on touch. The player or bot wins upon gaining six points.

Canvas Design

Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?

(0,0)



(400,400)

Code Design

Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

Concept	Purpose
Group	I want to make the power up icon look more appealing, so I want to use groups to design a more complex appearance.
onStep()	Use step events is crucial to making continuous motion and real time updates within the game. Pong requires all movement to be smooth, so onStep allowed the code to run so at a constant pace. Without it, smooth motion is impossible and the ball would not even move.
Def	I used Def multiple times in my code to organize it into smaller reusable parts. This helped me avoid repeating the same code many times and made my program much more organized and manageable.
App	I used App to store vital global variables for the game. It allowed me to control information for the entire game in one place instead of creating many variables to substitute, making it easier to control.

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Carnegie Mellon University
Computer Science Academy

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?
- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

One difficulty I faced was that the game will often end up in a loop where the bot cannot win, because the ball's speed did not reset and therefore the bot could not catch up with it. I tried to add additional code that reset the ball's speed, but the issue still occasionally recurred. Ultimately, I increased the bot's movement speed as well, and this problem almost never happened again. I also slightly improved the graphics of the game, using groups and gradients to add more depth.

Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose
Group	17	A Group is used to combine multiple shapes into a single object so they can be controlled together.	I used a Group to create a more complex power-up design instead of a single shape. This allowed me to move, hide, and detect collisions with the entire power-up as one object, making my code more efficient and visually interesting.
onStep()	55	The onStep function runs repeatedly to update the game over time, acting like a game	I used onStep to continuously move the ball and update all game elements such as collisions, shadows, and the bot movement. This allowed my game to feel

		loop.	smooth and animated instead of just what onKeyPress could offer
def	23, 26, 30, 39, 48, 55	The def keyword is used to define a function, which is a reusable block of code that performs a specific task.	In my program, I used functions like playerWins() and botWins() to handle scoring and resetting the game. This helped me avoid repeating code and made my program more organized and easier to update.
app	2, 3, 4	The app object is used to store global variables and control settings for the entire program.	In my code, I used app.counter to track time and control when the power-up should spawn. This allowed me to create timed events and manage game behavior more effectively in a centralized way.

Future goals

Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

If I had more time or were to restart this project, I would make some improvements to both my code and how I worked on it. For the code itself, I would focus more on improving the features I already have instead of just adding new ones. For example, I would make the power-up system more interesting with better timing and maybe different types of effects, and I would try to make the ball movement feel more natural instead of just bouncing in simple patterns. I would also clean up my code by using more functions so I do not repeat the same code as much.

Looking back at my process, I did not completely follow my original plan. I started with just basic Pong, but as I got more comfortable and confident, I added things like shadows and power-ups. These changes made the game more fun, but they also made things more complicated because I had to keep adjusting my code. If I did this again, I would plan those features earlier so my code is set up for them from the beginning.

For resources, I mostly relied on class notes and figuring things out on my own. This helped me understand what I was doing, but it also slowed me down at times. I think I could have used more tutorials or examples to learn faster ways to do certain things, especially for randomness and smoother movement.

In terms of time management, I spent a lot of time fixing small bugs and not enough time planning ahead. Next time, I would spend more time thinking through my design before coding, which would probably save time overall and make everything go more smoothly.

Overall, this project helped me learn not just how to code a game, but also how important planning, organization, and time management are when working on a bigger project.